

The Effects of Group Longevity on Project Communication and Performance

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This study investigated the communication behaviors and performances of 50 R&D project groups that varied in terms of group longevity, as measured by the average length of time project members had worked together. Analyses revealed that project groups became increasingly isolated from key information sources both within and outside their organizations with increasing stability in project membership. Such reductions in project communication were also shown to affect adversely the technical performance of project groups. Furthermore, variations in communication activities were more associated with the tenure composition of project groups than with the project tenures of individual engineers. These findings are presented and discussed in the more general terms of what happens in project groups with increasing group longevity.*

Group activities do not occur all at once or at a particular time; they take place over time. One of the major problems in the study of groups and project teams has been the general neglect of such a temporal perspective. Without an appreciation of what happens over time, answers to questions about how well groups are doing will be incomplete. More temporally based frameworks are needed to conceptualize the changes that are likely to occur in a group over time. The research reported here focuses on these issues by examining communication and performance differences among 50 R&D project groups whose professional members had been working together for different lengths of time.

THEORETICAL BACKGROUND

Communication Methods and Performance

To keep informed about relevant developments outside the organization as well as new requirements within the organization, project groups must collect and process information from a variety of outside sources (Thompson, 1967). The research of Menzel (1965), Allen (1977), and many others has consistently shown that interpersonal communications, rather than technical reports, publications, or other written documentation, are the primary means by which engineering professionals collect and transfer important new ideas and information into their project groups.

One explanation for this heavy reliance on personal contact lies in the strong distinction between science and technology. As discussed by Price (1965) and Allen (1977), most documentation for technological work is useful only when knowledgeable others are directly available to explain and supplement its written content. Engineers, therefore, can obtain information either through the very difficult task of decoding physical outputs and products or by relying on direct personal contact with other technologists. Their ability to rely on the written work is considerably less than that of the basic scientist. Thus, engineering groups keep abreast of their field and channel new technology and ideas into their work primarily through personal associations with other professionals.¹

Research on the normative effects of interpersonal communications has convincingly demonstrated that R&D project performance is strongly associated with high levels of technical

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In his research on engineering project teams, for example, Allen (1977) carefully demonstrated that only 11 percent of the sources of new ideas and information could be attributed to written media; the rest occurred through interpersonal communications.

communication by all project members with information sources within the organization (i.e., with high levels of internal communication). The empirical findings of Pelz and Andrews (1966) coupled with the longitudinal results of Allen (1970) and Farris (1969) strongly support the contention that stimulating direct communications between project group members and other internal professional colleagues can enhance project effectiveness. Surprisingly enough, however, in these studies contacts between project members and professional colleagues outside the organization (i.e., external communication) did not contribute to higher project performance, even though such organizational interchange was considered critical. According to Allen's (1977) explanation, most engineers are unable to communicate effectively with external professionals across organizational boundaries. A different communication method is needed for keeping up-to-date with technical advances outside the organization.

One way of dealing with this technology transfer problem is to rely on special boundary-spanning project members, identified and labelled gatekeepers by Allen and Cohen (1969). With the help of these key individuals, external information is channelled into project groups by means of a two-step communication process (Coleman, Katz, and Menzel, 1966). First, gatekeepers gather and understand external information; and second, they translate it into terms that are more meaningful and useful to their project colleagues. Gatekeepers, as a result, are extremely valuable, for they are the principal means by which external ideas and information are effectively transferred to project groups (Allen, 1977; Katz and Tushman, 1981).

Influence of Project Characteristics

Recent studies, however, suggest that not all R&D project groups are alike in the way they function or in the way they should be managed for effective communication (e.g., Whitley and Frost, 1973; Allen, Lee, and Tushman, 1980). The type of work that the project group does makes a great deal of difference. Engineers assigned to more research-oriented projects, for example, have very different information needs and, consequently, behave very differently from engineers concerned with product and process development projects. These, in turn, are quite different from groups involved in product modification and adaptation, i.e., technical service work. Definitions for these three project categories can be found in the methodology section.

Specifically, earlier research (Shilling and Bernard, 1964; Baker, Siegmann, and Rubenstein, 1967; Allen, 1977) has shown very clearly that for development and technical service projects, the best sources of technical information were within the organization. Development projects performed better when group members maintained higher levels of internal communication with other members of the R&D staff and with members of other organizational divisions, especially their clients in marketing and manufacturing. Similarly, internal communications among group members was important for the performances of technical service teams, particularly interaction between project members and their project managers (Allen, Lee, and Tushman, 1980). Increasing the direct external professional

contacts of development and technical service group members, on the other hand, has not been positively associated with higher project performance (Allen, Tushman, and Lee, 1979). Instead, technical gatekeepers were needed by these project groups to communicate effectively with professional sources outside the organization (Katz and Tushman, 1981). In sharp contrast, research projects were higher performing, not when group members maintained high levels of internal interaction, but when group members had strong direct communication with external professionals (Dewhirst, Arvey, and Brown, 1978; Katz and Tushman, 1979). In fact, research projects relying on a gatekeeping process for their external linkage were the least effective research groups.

One reason for this significant difference between research and development groups is that development projects are strongly local, in that their problems and solution strategies are defined and operationalized in terms of the particular strengths, interests, and experiences of the organizational subculture in which they are being addressed (Allen, 1977; Katz and Kahn, 1978). Development groups in different organizations may face similar problems, yet they define their solution approaches and parameters very differently. The coupling of bureaucratic interests and demands with such localized tasks and language schemes produces a communication boundary which makes it difficult for most development engineers to communicate effectively with outside professionals and consultants about their project-related activities (Lawrence and Lorsch, 1967; Tushman and Katz, 1980).

Research projects, on the other hand, are more universally defined, and consequently members of such projects are less impeded in their external communications across organizational boundaries. Research project members, as a result, are able to communicate directly and effectively with external R&D colleagues (Hagstrom, 1965). This is not to say that external information and new technological advances are unimportant for development projects. On the contrary, they are exceedingly important. What is implied by this explanation is that it becomes increasingly difficult for development engineers to mesh external ideas, suggestions, or problem solutions with internal technology that has become locally defined and constrained. Gatekeepers are needed to bridge the gap between localized development work and the larger technological world outside the organization.

Technical service projects are also local, but, unlike development projects, their members deal with more mature technologies and existing knowledge and products (Rosenbloom and Wolek, 1970). With such stable technologies, the specialized gatekeeper role is not as critical as in development work, for the managerial hierarchy can help keep technical service members sufficiently informed about external events and information through formal operating channels (Walsh and Baker, 1972; Tushman and Katz, 1980). Thus, the particular method by which R&D project groups effectively draw upon technological ideas and information outside the organization is strongly contingent on the nature of the project's work, even though direct member contact is an effective method for internal communication.

Group Longevity and Project Communication

Given the strategic importance of oral communication in all R&D project groups, one must carefully consider the effects of any social or group related processes that could significantly affect the interaction patterns of project members. Specifically, in the present research, the actual communication activities of project group members are examined as a function of the length of time the members have worked and shared experiences with one another, i.e., as a function of group longevity. As group longevity increases over time, a number of different but interrelated social processes begin to affect group behavior. Although these social processes are not part of the study's empirical design, it is argued here that they cause members of long-tenured groups to become increasingly isolated from outside sources of information and influence.

Behavioral stability. One of the more important principles in organizational theory is that groups strive to structure their work environments to reduce the amount of stress they must face by directing their activities toward a more workable and predictable level of certainty and clarity (Thompson, 1967; Pfeffer, 1981). Based on this perspective, project members interacting over a long period will develop standard work patterns that are familiar and comfortable, patterns in which routine and precedent play a relatively large part. Weick (1969), for example, discusses the strong tendency for groups to establish certain stable structures of interlocked behaviors and relationships simply because it keeps them feeling secure and confident in what they do. Furthermore, findings by Katz (1978a, 1979b) suggest that group members who have been performing their jobs for extended time periods are relatively less responsive to the challenging aspects of their project activities. Such individuals come to rely more and more on their customary ways of doing things to complete their everyday project requirements.

It is posited that one of the consequences of such behavioral stability is that long-tenured project groups become increasingly isolated from outside sources of relevant information and important new ideas (Pelz and Andrews, 1966; Dubin, 1972). With increasing group longevity, project members gradually become less receptive toward communications that threaten to disrupt significantly their comfortable and predictable work practices and patterns of behavior (Staw, 1977; Katz, 1980). The extent to which such groups are willing or even feel they need to expose themselves to new information lessens with time. Thus, instead of being vigilant in seeking information from outside sources, they become increasingly complacent about external events and new technological developments.

Selective exposure and group homogeneity. Another set of forces affecting the amount of outside contact for projects with longstanding membership is the tendency for group members to communicate only with those whose ideas and viewpoints are in accord with their own interests and current perspectives. One of the main principles of human communication, often referred to as selective exposure (Rogers and Shoemaker, 1971), is the strong tendency for individuals to communicate with others who are most like themselves or who are most likely to agree with them. Over time, project members learn to interact selectively to avoid messages and information that

might conflict with their established practices and dispositions, thereby reducing their overall levels of outside contact.²

It is further argued that as group members continue to interact and build a history with one another, a more homogeneous set of understandings about the group and its environment will develop through informational social influence (Homans, 1961; Berger and Luckman, 1966; Salancik and Pfeffer, 1978). Group homogeneity can come either from similarity of social backgrounds and characteristics or from group members remaining in their project positions long enough to make shared socialization and shared group experiences a meaningful basis of mutual support (Grusky, 1964; Kanter, 1977). Such shared meanings not only provide group members with a strong sense of identity but also demarcate the group from other entities both within and outside the organization, leading to further reductions in the group's overall level of outside contact. In short, it is argued that with increasing group longevity, the effects of behavioral stability, selective exposure, and group homogeneity combine to reduce the group's willingness to search out and actively internalize new or conflicting knowledge and developments. Accordingly, the following is hypothesized:

Hypothesis 1. Project groups characterized by high levels of group longevity will have significantly less communication with relevant information sources outside their project groups.

Role differentiation. As project members work and gain experience with one another over a long period, group members become increasingly specialized in their particular problem areas and project assignments, resulting in greater role differentiation and less overall interaction within the group (Porter, Lawler, and Hackman, 1975). Gradually, they come to know each other well, know what to expect from each other, and, consequently, there is simply less need for talk and interaction among all project members. Their perceptions of each other's capabilities, interests, and contributions become clearer and more well defined, with members functioning in ways that gradually regularize and stabilize these role differences (Bales, 1955). As a result, overall levels of intragroup interaction will be less with high group longevity, causing groups to lose access to much of their internal talent and reducing their total capacity for learning new ideas from one another. Thus, the following hypothesis is put forth:

Hypothesis 2. Project groups characterized by high levels of group longevity will have significantly reduced levels of intraproject communication.

Group Longevity, Performance, and Project Communication

Reduced levels of project communication can have very serious consequences, especially for R&D groups, given their strong dependence on external ideas and developments as well as their need for effective coordination with other organizational areas, particularly manufacturing and marketing (Utterback, 1974; von Hippel, 1978). Since project members rarely have all the requisite work-related knowledge and expertise to accomplish their projects successfully, information and consulting support must be gathered from many sources outside the project.³ Given the critical importance of both internal and external communications for project performance and the

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Under such circumstances, even the outside information that is processed may not be viewed in the most open and unbiased fashion. Janis and Mann (1977), for example, discussed at great length the many cognitive defenses and distortions commonly used to support and protect particular decisional policies and strategies.

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The extent to which increased isolation may be a severe problem depends considerably on the nature of the project's work. Groups working on routine problems or working with constant technologies may not suffer as a result of less external vigilance, for internal expertise and experience are sufficient. Generally speaking, as groups function in a more changing technological environment and work on tasks requiring greater levels of creativity and innovativeness, the effects of increased isolation are more likely to be problematic.

effects that various social processes have been argued to have on project member interactions as group longevity increases, one should also find a significant relationship between project performance and group longevity, as has been found in three previous studies. Shepard (1956) was the first to relate performance to group longevity or mean project tenure (calculated by averaging the individual project tenures of all project members). In his small sample of R&D groups, performance increased up to about 16 months average tenure but thereafter decreased. Pelz and Andrews (1966) and Smith (1970), found similar curvilinear relations between mean group tenure and performance — the “optimum” group longevity mix being between three and four years.

The present study also focused on the relationship between group longevity and R&D project performance. This time, however, the research examined clearly defined project groups, used direct measures of project performance, and most important, tried to explain the curvilinear association in terms of varying levels of project member communication. The argument that group longevity is related to performance through project communication rests on the assumption that communication directly influences performance. Although the opposite assumption is also probably valid to some extent, there has been considerable longitudinal research, as previously cited, to demonstrate the strong positive effects of communication on technical performance. The basic model (Figure 1) underlying this study, therefore, is that group longevity relates to project performance primarily through shifts in project communication produced by the various social processes that underlie the longevity characteristics of the group.

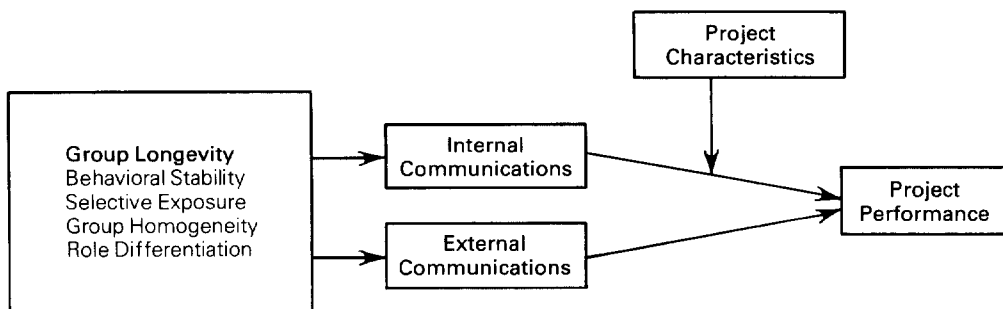


Figure 1. A model of the relationships among group longevity, communication, and project performance.

Specifically, as group longevity increases, the social processes lead to lower levels of project communication which, in turn, account for or mediate the relationship between group longevity and project performance. Most likely, project communication will mediate the group longevity/performance relationship only when direct project member interaction is appropriate for more effective technical performance. Thus, the following hypotheses are proposed with respect to the internal and external communications of the three different categories of R&D project work.

Hypothesis 3. Reduced levels of internal communication will mediate the relationship between group longevity and project performance only for development and technical service groups.

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Hypothesis 4. Reduced levels of external communication will mediate the relationship between group longevity and project performance only for research project groups.

Since alternative methods for effective external communication have been shown to be more suitable than direct project member interaction in situations of development and technical service work, hypothesis 5 is also proposed with respect to external contacts.

Hypothesis 5. Reduced levels of external communication will *not* mediate the relationship between group longevity and project performance for development and technical service project groups.

METHODOLOGY

Research Setting

This study was carried out at the R&D facility of a large American corporation. Geographically isolated from the rest of the organization, the facility employed a total of 345 engineering and scientific professionals, all of whom participated in the study. The laboratory was organized into seven departmental laboratories which, in turn, were organized into 61 separate project groups, ranging in size from 3 to 15 members. These project groups remained stable over the course of the study, and each professional was a member of only one group. To conduct a study involving group longevity, one cannot readily rely on sampling techniques to derive group measures; it is critical that all project members be carefully identified and that relevant tenure, age, and communication information be collected from as many project members as possible. In the present study, a project group was included in the analyses as long as appropriate scores of tenure and age were obtained from over 75 percent of all project group members. Based on this cutoff, complete data were successfully obtained on a total of 50 project groups, representing 82 percent of all project groups within this R&D facility. Table 1 presents the means and intercorrelations of the main variables measured in this study. As shown in this table, project size was not related to any of the main variables.

Finally, one must be cautious when interpreting the results of this study, since the data represent only a cross-sectional view

Table 1

Means and Intercorrelations of Variables*

Variables	1	2	3	4	5	6	7	8	9	10	11
1. Intraproject communication											
2. Departmental communication	.12										
3. Laboratory communication	-.06	.31									
4. Organizational communication	.10	.05	-.13								
5. Professional communication	.18	.05	.07	-.07							
6. Operational communication	-.03	-.07	-.13	.17	.00						
7. Project performance	.11	-.17	-.10	.14	-.06	-.12					
8. Group longevity	-.14	.10	-.06	-.13	-.14	.12	-.16				
9. Mean project age	-.10	.01	.11	-.07	-.20	.15	-.17	.47			
10. Mean organizational tenure	-.11	.11	.15	-.03	-.14	-.06	-.13	.61	.65		
11. Project size	.23	-.03	-.15	-.07	.20	-.14	.06	-.09	-.02	-.07	
Mean	45.4	17.3	10.4	22.8	1.27	4.61	4.59	3.41	38.0	7.04	6.46

*With $N = 50$ projects, a correlation of 0.24 is significant at the .05 level.

of what has been presented as essentially longitudinal phenomena. Strictly speaking, a longitudinal design is necessary to test suppositions of a temporal or developmental nature, although a cross-sectional investigation can serve as an exploratory first step — one directed at giving enough face validity to the hypotheses to warrant investing in future longitudinal studies.

Tenure and Demographic Data

Demographic data were collected from all R&D professionals, including their age, educational degrees, and the number of years and months they had been associated with their specific project group and with the overall laboratory facility. As in previous studies (Shepard, 1956; Pelz and Andrews, 1966; Smith, 1970), group longevity or mean project tenure was calculated by averaging the individual project tenures of all project members. The mean was used to obtain a representative estimate of how long project members had worked together and shared experiences with one another. It is important to recognize, then, that group longevity is not the length of time the project has been in existence, but the average length of time project members have interacted with each other. Thus, the measure of group longevity is not tied to project phase, nor is it necessarily related to how long R&D professionals have been working in that particular problem area within the company. Most of the facility's projects, however, were of long duration, having been organized around specific, long-term types of problems in areas such as fiber-forming development, urethane development, and yarn technology. Finally, measures of mean age and mean organizational tenure were also calculated for each project group by averaging the chronological ages and laboratory tenures reported by all group members.

Project Group Characteristics

In R&D settings, tasks can differ along several dimensions, including specific versus general problem-solving orientation, and generation of new knowledge versus utilization of existing knowledge and experience (Rosenbloom and Wolek, 1970). Based on these dimensions, the following project categories were developed with the help of the managers of the laboratory:

1. *Basic research*. Work intended to apply to a broad range of applications or to the development of new knowledge about an area.
2. *Applied research*. Work involving basic knowledge for the solution of a particular problem; the creation and evaluation of new concepts or components, but not development for operational use.
3. *Development*. Combination of existing feasible concepts, perhaps with new knowledge, to provide a distinctly new product or process; application of known facts and theory to solve a particular problem through exploratory study, design, and testing of new components or systems.
4. *Technical service*. Cost and performance improvement of existing products, processes, or systems; recombination, modification, and testing of systems using existing knowledge; opening new markets for existing products.

Using these definitions, respondents selected the category that best characterized the objectives of their project and indicated, on a three-point scale, how completely the project's objectives were represented by the selected category. The twelve possible answers were scored along a single scale ranging from completely basic research to completely technical service. As in Pelz and Andrews (1966), respondents also indicated what percentage of their project's work fell into each of the four categories, and a weighted average of the percentages was calculated for each respondent (Tushman, 1977). The scored responses to these two questions were then averaged (Spearman-Brown reliability = .91).

By pooling members' responses to obtain project scores, a project could readily be identified as being predominantly either: (1) research (a combination of basic and applied research categories), (2) development, or (3) technical service. Analysis of variance was used, as described by Katz and Tushman (1979), to ensure the appropriateness of combining individuals' perceptions of their activities for the aggregate categorization of each particular project group.

Technical Communication

To measure actual communications, professionals kept a record (on specially prepared lists) of all other professionals, both within and outside the organization, with whom they had work-related, oral communication for a given sampling day (including to whom they talked and how many times they talked with that person during the day). These sociometric data were collected on a randomly chosen day each week for 15 weeks. Sampling was designed to allow each weekday to be represented an equal number of times. Respondents did not report contacts that were strictly social, nor did they report written communications. These research procedures were similar to those used in other sociometric communication studies, including Allen and Cohen (1969) and Whitley and Frost (1973).

During the 15 weeks, the overall response rate was 93 percent. Moreover, 68 percent of all reported communications in the laboratory were mentioned reciprocally by both parties. Given these high rates of response and mutual agreement (see Weiss and Jacobson, 1960, for comparative data), these methods provided a very accurate log of the oral communications of the professionals in the laboratory.

Project communication was a measure of the average amount of technical communication per person per project over the fifteen weeks. Following Katz and Tushman (1979), six mutually exclusive communication measures were operationalized for the communication domain of each project group as follows:

1. *Intraproject*: Communication reported among all members of the project group.
2. *Departmental*: Communication reported between project members and other R&D professionals within the same functional department.
3. *Laboratory*: Communication reported between project members and R&D professionals outside their functional department but within the R&D facility.

4. *Organizational*: Communication reported by project members with other individuals outside the R&D facility but within other corporate divisions, such as marketing and manufacturing.

5. *Professional*: Communication reported by project members with professionals outside the parent organization, including professionals in universities, consulting firms, and professional societies.

6. *Operational*: Communication reported by project members with external operational areas, including vendors and suppliers.

The first four were measures of internal communication, the last two of external communication.

Communication measures for these six independent domains were calculated by summing the relevant number of interactions reported during the 15 weeks, with appropriate normalizing to adjust for the number of project team members, following the methods of Katz and Tushman (1979). Although the overall response rate was extremely high, raw communication data of respondents with missing data were proportionately normalized to adjust for the number of missing weeks. Finally, none of the six measures of communication were significantly intercorrelated at the .10 level of significance, with the exception of departmental and laboratory communication ($r = .31$; $p < .05$).

Project Performance

Since comparable measures of project performance have yet to be developed across different technologies, a subjective measure, similar to that used by Lawrence and Lorsch (1967), was employed. Each department manager ($N = 7$) and laboratory director ($N = 2$) was separately interviewed and asked to evaluate the overall technical performance of all projects with which he was technically familiar. He was asked to make informed judgments based on his knowledge of and experience with the various projects. If he could not make an informed judgment for a particular project, he was asked not to rate the project. Criteria the managers considered (but were not limited to) included: schedule, budget and cost performance, innovativeness, adaptability, and the ability to cooperate with other groups in the organization. Each project was independently rated by an average of five managers on a seven-point scale (from very low to very high). As the performance ratings across the nine judges were highly intercorrelated (Spearman-Brown reliability = .81), individual ratings were averaged to yield overall project performance scores. In making these informed judgments, the managers were not cued about the purpose of this study. Furthermore, post-study conversations revealed that none of the managers could readily identify projects with high levels of group longevity.

Technological Environment and Project Interdependence

Since perceived environmental variability and project interdependence have been related to project communication in previous studies (Allen, 1977; Tushman, 1977), measures for these two concepts were collected to check for possible alternative explanations in analyses involving communication

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differences. Perceptions of environmental change were measured by asking respondents to answer the following question on a five-point scale ranging from 1 ("to a very little extent") to 5 ("to a very great extent"):

We are interested in how rapidly you see the demands of your job changing. To what extent are the techniques or skills or information needed for your project work changing?

Project interdependence reflects the extent to which one's project assignments require working with other groups in the organization. Following Tushman (1977), respondents indicated (on a ten-point scale) the degree to which they had to work (1) with other project members, (2) with departmental colleagues outside the project group, and (3) with individuals from marketing, manufacturing, and other organizational divisions.

Since projects are the units of analysis, the homogeneity of project members' perceptions of these environmental and interdependent variables were tested to check for the appropriateness of pooling, using the same methods described in studying project group characteristics. As pooling was appropriate for each variable, individual responses were combined to obtain project scores.

RESULTS

Project Performance

The 50 projects had group longevity scores ranging from several months to almost 13 years, with an overall sample mean of 3.41 years. The mean rating of project performance, as provided by the evaluators, ranged from a low of 3.0 to a high of 6.4, with a mean of 4.59. When project performance was examined as a function of the mean project tenure of group members, it appeared to be highest in the 2- to 4-year interval, with lower performance scores both before and after.

To get a better idea of whether a clear pattern might emerge between group longevity and project performance, the original performance data were subjected to a smoothing technique using Tukey's (1977) recently developed 3RSSH smoothing procedure. As suggested by Tukey, this technique was not used to test any specific hypothesis but was applied simply to get a more meaningful picture of the data in order to use more appropriate statistical analyses and comparisons. The resultant calculations, plotted in Figure 2, illustrate very clearly that performance was highest for projects with a mean group tenure of between two and four years. More interestingly, the shape of these smoothed data points indicate that performance might begin and continue to decline for project groups whose members have averaged five or more years of work in their particular project groups. Given this exploratory pattern, more confirmatory statistical analyses can be used to test relationships using only the original raw measures of performance and group longevity.

To corroborate any pattern of significant differences in the distribution of actual project performance as a function of group longevity, the 50 groups were divided into separate mean project tenure categories based on the curvilinear shape shown in Figure 2. There seemed to be at least three different tenure periods represented in the data: 0 to 1.5 years, 1.5 to 4.9 years,

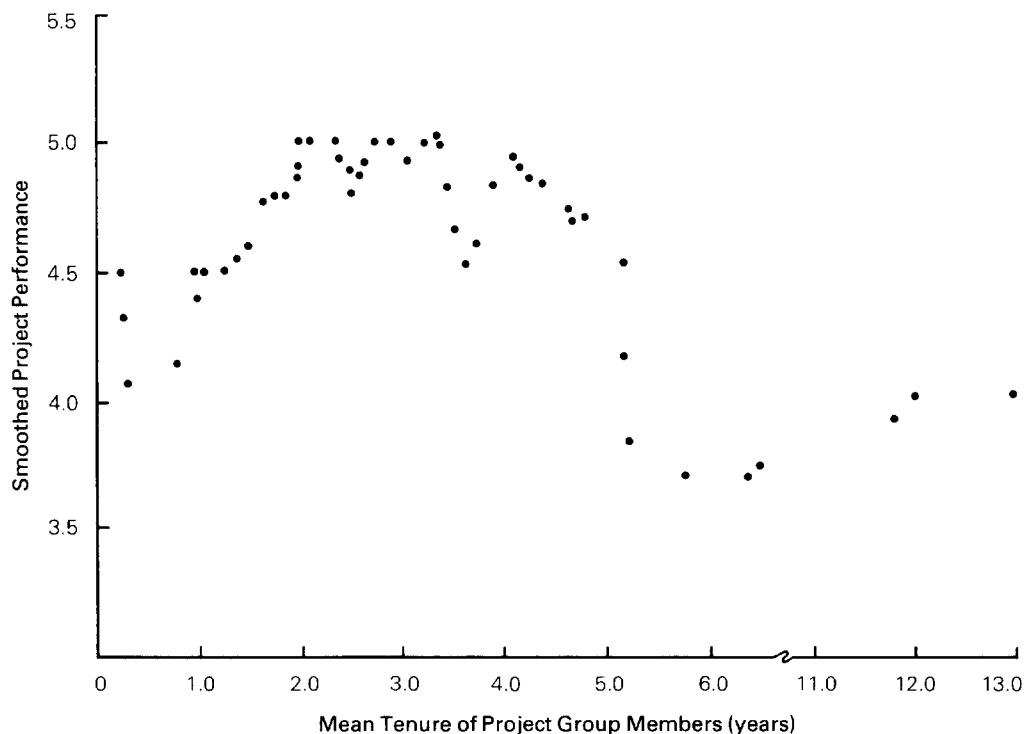


Figure 2. Smoothed project performance as a function of the mean tenure of project group members.

and 5 or more years. For additional exploratory purposes, the 30 project groups falling in the middle or medium tenure range were further subdivided into three equal categories as shown in Table 2. The first interval of 0 to 1.5 years corresponds to an initial learning or building phase that matches very well with the curvilinear performance findings of the previously cited studies. Similarly, the last category of project groups (5 or more years) corresponds exactly with the performance decline found in previous research, as well as with the time interval commonly used to estimate the "half-life" of technical information (Dubin, 1972).

Table 2 shows that performance was significantly lower for projects with group longevities of less than 1.5 years or more

Table 2

Project Performance as a Function of Group Longevity

Variables	Categories of Group Longevity (in years)					All Project Groups (N=50)
	0-1.5 (N=10)	1.5-2.5 (N=10)	2.5-3.5 (N=10)	3.5-5.0 (N=10)	5.0 or more (N=10)	
Mean project performance*	4.29	4.89	4.87	4.82	4.07	4.59
Standard deviations	.99	.67	.70	.59	.52	.76

*Based on a 1-way ANOVA test, the mean project performance scores are significantly different across the five group longevity categories; $F(4,45)=2.89$; $p<.05$.

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than 5 years and was significantly higher across all three middle tenure categories. Thus, the findings strongly support the curvilinear association between project performance and mean project tenure within this organization.

Group Longevity or Mean Age of Project Members

As expected, projects with higher group longevity were also staffed by older engineers ($r = .47$, see Table 1). Of course, this raises the possibility that the decrease in performance associated with high group longevity had little to do with the group itself. It may have resulted, instead, from individuals becoming less current in their technical fields as they aged. Furthermore, a third variable, mean organizational tenure of project members, was also strongly correlated with group longevity and group member age and, as a result, was included in the next comparative analysis.

To examine the relative importance of these three variables, successive nonlinear regressions were run on project performance to compare the additive effects of each variable after controlling for the effects of the other two. The first row of results in Table 3 confirms the significant curvilinear (concave downward) relationship between project performance and group longevity. The quadratic regression equation containing both mean project tenure and mean project tenure squared accounted for significantly more variance in project performance ($Ra = .29$) than the simple linear regression containing only mean project tenure ($Rc = .16$).

The remaining three regression analyses reported in Table 3 demonstrate more convincingly that it is tenure in the project team and not age or organizational tenure that is more likely to influence project performance. As shown by the significant multiple R difference between the two regression equations in the second row, the addition of the two group longevity terms to the regression already containing the linear and quadratic terms of mean age and mean organizational tenure was still able to explain significant amounts of additional performance data in the 50 project teams. The findings in the third and fourth rows, on the other hand, indicate that neither mean age nor mean organizational tenure showed any further significant association with project performance after controlling for the effects of the other two variables.

Table 3

Comparison of Multiple Regressions on Project Performance by Mean Project Tenure, Mean Age, and Mean Organizational Tenure

Control variables in regression equation	Variables added to regression equation	Multiple correlations with performance		F-values for $Ra - Rc$
Rc	Ra	Rc	Ra	
$Rc = Rm$	$Ra = Rmp$	.16	.29	3.01*
$Rc = Rmo + ma$	$Ra = Rc + mp$	.24	.39	2.48*
$Rc = Rmo + mp$	$Ra = Rc + ma$	.32	.39	1.52
$Rc = Rmp + ma$	$Ra = Rc + mo$	.36	.39	0.86

* $p < .10$; $N = 50$

Note: In testing for significant additive differences between Rc and Ra , the following definitions are used: m = mean project tenure of project members; mp = mean project tenure of project members + mean project tenure squared; ma = mean age of project members + mean age squared; mo = mean organizational tenure of project members + mean organizational tenure squared.

Group longevity, however, is only a central tendency measure; one must also investigate individual member variations about the group's mean tenure. Clearly, ongoing staffing decisions will create markedly different distributions of project tenure within groups, even though their group longevity scores turn out to be the same. As a result, the standard deviation of project tenures across group members was used as one measure of the distribution about the project mean to explore its relationship with project performance. At first, a distinct curvilinear relation between performance and the standard deviation of project tenure was also found. Closer scrutiny, however, revealed that this significant association probably derived from the very high relation between the measures of mean project tenure and their standard deviations ($r = .51$). As a result, coefficients of variation were used to reexamine the performance/variance association, using Tukey's (1977) smoothing procedure. No clear relation between project performance and this measure of project tenure variance could be detected across the 50 projects. Nor did any obvious relationship emerge within any of the project tenure categories defined in Table 1. Further research, therefore, is probably needed to explore the effects of different tenure distributions where the relevant variance measures are based on more conceptual criteria such as differences between the individual project tenures of managers and subordinates, newcomers and veterans, or between members of different disciplines or technical specialties.⁴

Project Communication

Having demonstrated a strong association between group longevity and the technical performances of the 50 R&D project groups within the current site, we can now examine the various communication factors that might be affecting performance as group membership ages and stabilizes. It was hypothesized (hypotheses 1 and 2) that work-related communications to sources of information outside the project group and among group members themselves would be significantly lower for high longevity teams. Thus, the covariation between low project performance and high group longevity might be explained through low levels of project interaction with certain internal and external information domains, depending on the nature of the project's work.

To investigate these proposed relationships empirically, we first tested for significant differences in the actual communication behaviors of project groups to each of the six communication domains described in the methodology section as a function of group longevity. Significant variations were discovered for the two internal measures of intraproject and organizational communication and for the measure of external professional communication. The other communication measures revealed no strong differences across the five tenure categories.

Table 4 shows the significant variations in actual interaction for the three communication measures across the five categories of group longevity. In support of hypothesis 1, contacts outside the project group were significantly lower for high longevity teams; in fact, they varied curvilinearly with group longevity in a pattern congruent to that of project performance. Contacts with individuals from other organizational divisions and contacts with

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Since individual project tenures are not likely to be normally distributed within a project, the coefficient of variation is not without its problems. As a result, other measures could be used. The most promising ones would probably involve some kind of clustering or modal analysis to distinguish between distributions that are unimodal, bimodal, etc. Such measures, however, are extremely hard to compare across projects of different sizes and are also very sensitive to particular cutoff parameters. Considering these difficulties and the robustness of metric techniques vis-à-vis the normal distribution, variance parameters defined around a normal distribution, including the standard deviation, coefficient of variation, skew coefficient, and kurtosis, still seem most appropriate. In the present study, all of these measures were used to explore the performance/variance relationship but without any additional success.

Table 4

Mean Communication Frequencies as a Function of Group Longevity

Areas of communication (means)	Categories of group longevity (in years)					All project groups (N=50)
	0-1.5 (N=10)	1.5-2.5 (N=10)	2.5-3.5 (N=10)	3.5-5.0 (N=10)	5.0 or more (N=10)	
Intraproject communications	42.0**	101.0**	110.0**	180.0**	69.0**	100.0
Organizational communications (per person per month)	17.5*	20.3*	30.0*	25.6*	20.4*	22.8
External professional communications (per person per month)	.81*	.98*	2.04*	1.83*	0.69*	1.27

* $p < .10$; ** $p < .05$

Note: A 1-way ANOVA was used to test for significant mean differences across the five group longevity categories.

Because intraproject communication frequencies had to be adjusted for the number of possible interactions (Katz and Tushman, 1979), intraproject communication scores cannot be linked to an absolute scale. To show relative intraproject differences across the various categories, however, the intraproject measures have been standardized to an overall sample mean of 100.

external R&D professionals increased within the initial range of mean group tenure but were then significantly lower with high levels of group longevity. The second hypothesis is also supported, in that members of high tenured project groups had significantly less interactions among themselves. Thus, with high group longevity, R&D projects were not only more isolated from outside sources of information and influence, but they may also have been more stabilized in their individual project roles and contributions. To illustrate these parallel variations more clearly, the communication and project performance variables are plotted as a function of group longevity in Figure 3.

Project Performance, Communication, and Group Longevity

Given the significant differences in the internal and external communications of project teams along the group longevity continuum, the next question is whether such differences accounted for the lower performance ratings of the long-tenured groups. To investigate the remaining hypotheses, regression analyses were run on project performance to test for the additive effects of group longevity and group longevity squared after controlling for the effects of project communication. Furthermore, since research, development, and technical service project groups differ significantly in the way they effectively communicate externally, these regression analyses were run separately for each of them.

Table 5 indicates that group longevity did not reveal any additional significant association with project performance after partialling out the effects of communication behavior. There were no significant changes in the multiple R 's for either research, development, or technical service project groups, although the most important area for project communication did differ significantly across the three project categories. These regression analyses, then, suggest that group longevity affects project performance, at least in part, by operating

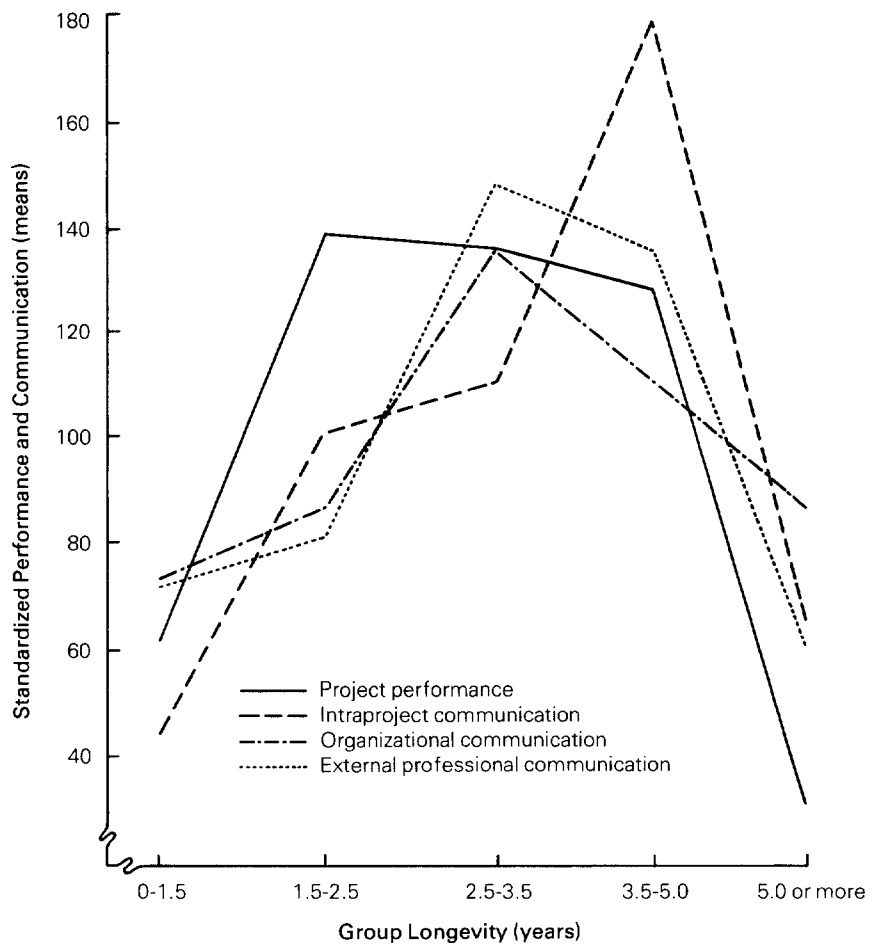


Figure 3. Standardized performance and communication means as a function of group longevity.

Table 5

Multiple Regressions on Project Performance by Group Longevity Terms After Controlling for Project Communication

Project type	Communication variables controlled for in regression equation R_c	Multiple Correlations		F -values for $(R_a - R_c)$
		R_c	R_a plus group longevity terms	
Research ($N = 12$)	Intraproject Organizational External Professional**	.61	.65	.26 (N.S.)
Development ($N = 21$)	Intraproject Organizational** External Professional	.47	.58	1.31 (N.S.)
Technical service ($N = 17$)	Intraproject* Organizational External Professional	.38	.56	1.36 (N.S.)

* $p < .10$ and ** $p < .05$ indicate communication variables with significant regression coefficients.

N.S. = No significant difference in project performance after adding group longevity plus group longevity squared.

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through reductions in the direct communications of all project members to particular information domains.

To test hypotheses 3, 4, and 5 directly and gain additional insight into the preceding regressions, Table 6 shows a reexamination of the long tenured part of the relationship between performance and mean project tenure while controlling for the effects of communication. This was done by dividing the sample at approximately the median level of group longevity.⁵ As shown by the simple correlations in the first row of Table 6, projects in all three R&D categories revealed a significant decrease in project performance with increasing group longevity. With one exception, moreover, the remaining correlations also showed a strong tendency for group members in all three project categories to interact less often with individuals from the three communication domains. For each project type, however, the isolation from a certain key area was particularly strong. There was an obvious decrease with group longevity in the external communication of research project groups, in the communication between development projects and other organizational divisions, and in the intraproject communication of technical service teams.

Finally, the partial correlational findings in Table 6 strongly support hypothesis 3. For development and technical service

Table 6

Partial Correlations between Group Longevity and Project Performance and Communication for Projects with Mean Tenure of at Least 2.5 Years

Variables correlated with group longevity	Type of Project		
	Research (N=6)	Development (N=12)	Technical service (N=12)
Correlations			
Project performance	-.62*	-.39*	-.44*
Internal communications			
Intraproject communication	-.26	-.14	-.72***
Organizational communication	.27	-.53**	-.12
External Communications			
External professional communication	-.51	-.23	-.38
Partial correlations			
Project performance controlling for internal communications	I*	-.18	-.12
Project performance controlling for external communications	I	-.42*	-.36

* $p < .10$; ** $p < .05$; *** $p < .01$

*I = Insufficient number of projects for partial analyses.

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The median level of group longevity was 2.6 years. However, since five project groups clustered extremely close to this median level, the sample was actually split at 2.5 years in order to increase the sample sizes in the analyses.

projects, direct measures of internal communication covaried sufficiently with both performance and group longevity to mediate their negative association. The data also supported hypothesis 5, in that reduced levels of direct external professional communication by all project members did not account for the lower performances of long-tenured development and technical service groups. Unfortunately, hypothesis 4 could not be tested as explicitly since there were not enough long-tenured research teams in the organization studied, although the regression findings in Table 5 suggest that lower amounts of

external communication probably accounted for the lower performances of research projects with high group longevity. Despite this omission, these findings clearly support the important mediating role of communications in explaining the significant variation of project performance across the group longevity categories.

Technological Environment and Project Interdependence

Project communication patterns are significantly affected by how interdependent the project's work is with other areas. Furthermore, the need for keeping abreast of external technological developments through outside professional communication is also affected by how rapidly the relevant technological environment is changing. One possible explanation for the significant reduction in communication behavior with high group longevity, then, is that these project groups also differed with respect to their projects' interdependence with other organizational areas or with respect to the changing nature of their relevant technological environments. In checking for these possibilities, no significant mean differences emerged for these variables across the categories of group longevity either for the sample as a whole or for each type of R&D activity. Thus, members comprising long-tenured project groups interacted less often among themselves, less often with other divisional areas of the company, and less often with outside R&D professionals than members in the medium range of group longevity, even though perceptually they were just as interdependent and their technological environments were changing just as rapidly.

DISCUSSION

The thrust of these findings emphasizes the important role that group longevity and its underlying social processes can play in influencing the information processing behaviors of project group members which, in turn, affect project performance as indicated by the curvilinear relationship between performance and mean project tenure. This relationship, moreover, was present independent of age and organizational tenure. Thus, performance appears to be significantly affected by the tenure composition of the particular group currently assigned to the project effort. This result is particularly important in light of the many suppositions regarding the general decline of technical performance as engineers age and become less current some-time after their mid-thirties.

The upward slope in performance probably reflects the positive effects of learning and team building as new project members contribute fresh ideas and approaches while also developing a better understanding of each other's capabilities, of the technologies involved, and of their working relationships. Such positive effects, however, appear to taper off for teams whose members have continued to work together for a long period of time. The performances of long-tenured project groups were significantly lower in all three areas of R&D effort, including research, development, and technical service.⁶

Certainly it is possible that, on the average, long-tenured project groups come to be staffed by less technically competent or perhaps less motivated engineers, although the average project tenure of supervisors from the ten long-tenured teams did

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From a historical perspective, individuals may have been working in a particular project or work area for a long period of time, but the collection of technologists currently assigned to the project could fall in any of the group longevity categories. Development work on one project, for example, had been going on for many years, but the group currently assigned had an average project tenure of only 2.3 years. Clearly, the behaviors and performances of project groups are also affected by the particular phase of the project, including preliminary design, laboratory testing, and product transfer. Although it was not possible to control or test for the importance of project phase, future research should address the interactions between group longevity and project phase.

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not significantly differ from the project tenures of the 30 supervisors in the medium range of group longevity. Nor were there any significant differences in overall educational levels, technical reports written, or in the number of professional journals read (at least on a self-report basis).

More important, recent follow-up visits to this facility showed that the same proportion of professionals from both the long- and medium-tenured project groups had been promoted to higher level managerial positions above the project leadership-level during the five-year interval since the collection of the original data. Of the engineers who had been working on projects with a group longevity of 1.5 to 5 years, 15 percent attained higher level managerial positions. The comparable percentage for engineers working in the ten long-tenured projects was 13 percent. In addition to these managerial advances, shortly after the data collection, the company introduced a dual ladder promotional system to reward individual professionals whose "technical competency and contributions are well-recognized," and 12 percent of the engineers from the medium-tenured project teams were promoted to equivalent managerial positions on the technical side of the dual ladder. Surprisingly enough, the comparable percentage for the long-tenured teams was slightly higher, about 19 percent. Such promotional developments indicate that neither individual competence nor perhaps the importance or visibility of the project can account for the significant difference in technical performance between medium- and long-tenured project groups.

As hypothesized, it was the clear reductions in the direct communications of all project members to key information sources that were able to account for these performance differences. For projects with high group longevity, group members were communicating less often among themselves, with individuals from other organizational divisions, and with external professionals. Since the discussion and transfer of technical information and new ideas is important to effective project performance in R&D settings, it seems reasonable to attribute the lower technical performance of long-tenured groups, at least in part, to these differences in information processing activity.

One should also emphasize that it is not a reduction in project communication itself that leads to a decrease in performance. Indeed, some of the measures of project communication did not diminish significantly with higher levels of group longevity. Rather, a decline in performance is more likely to arise from the tendency of group members to ignore and become increasingly isolated from sources that provide the most critical kinds of evaluation, information, and feedback. Since research, development, and technical service projects differ significantly in these information sources as well as in the methods by which such information can be effectively obtained and processed, project groups in all three work categories are likely to suffer more when there is widespread isolation from their most critical information domains. Thus, performance is lower when research members fail to pay sufficient attention to events and information in their external R&D community or when development and technical service project members fail to communicate with their internal R&D colleagues or with their client groups from marketing and manufacturing.

Since development project performance is more sensitive to a gatekeeping communication method and less sensitive to the external interactions of all project members, less external professional contact may not have affected the performance of development groups directly. Nevertheless, reduced levels of internal project communication could have hindered the extent to which appropriate project members were able to complete the gatekeeping role, since it depends on considerable internal interaction. Although this issue cannot be addressed with the present data, it is worth noting that none of the five high-longevity development groups had a project member fulfilling the gatekeeping role. From a managerial viewpoint, it would be important to determine whether a gatekeeping project member could have maintained high project performance. In addition, could the assignment of a technical gatekeeper to a long-tenured development group facilitate high performance, or might group members collude to ignore the suggestions and contributions of such an individual? While a great deal has been learned about the gatekeeping and other critical role functions for product and process innovations (Allen, 1977; Katz and Tushman, 1981; Roberts and Fushfeld, 1982), considerably more must be learned about how these roles are influenced by increasing stability in both group and laboratory memberships.

Finally, the proposed model of relating group longevity to project performance through project communication is only one possible sequence of these three variables. One could also argue that as groups communicate less and become lower performing, the organization pays less attention to them, thereby allowing these low performing groups to accumulate greater levels of group longevity. While this alternative sequence is possible, the members of projects with high group longevity certainly did not view themselves as less connected to other organizational areas, nor did their subsequent promotional rates suggest that the organization was neglecting them. Furthermore, when we discussed these findings with a number of R&D managers, they usually agreed that they would be more likely to "shake things up" in low-performing groups, at least in terms of shifting personnel, preferring to leave high-performing groups alone for fear of disrupting their performance, although several exceptions were called to our attention.

Implications

What is also important to recognize from this study is that communication behaviors of engineers in comparable tenured groups were more similar to the behaviors of their fellow group members than to the communication patterns of engineers working in groups from different tenure categories. In the laboratory studied, for example, there were no clear trends in any of the communication activities of individual engineers when they were examined as a function of individual project tenure. Only when engineers were grouped according to their project affiliations were there clear and obvious reductions in communication as a function of high group longevity. Engineers with high levels of project longevity, therefore, did not necessarily have less communication if they were not part of a long-tenured project group. In contrast to individual-type explanations, such findings strongly support the important influence of the social context in mediating individual behaviors and activities. While many other studies have shown the direct

effects of the social environment on job attitudes and perceptions, few studies have demonstrated such effects on work-related behaviors. The fact that communication behaviors were more associated with group characteristics than with individual ones underscores the argument that perceptions and responses do not take place in a social vacuum but develop through successive encounters with work and group environments (Katz and Van Maanen, 1977; Salancik and Pfeffer, 1978). Thus, one must carefully consider the social context in which project activities are being carried out in order to understand more fully how group members define and interpret their work experiences and to gain a more complete picture of group behavior.

The behavioral patterns reported in this study are compatible with the viewpoint that, over time, members of project groups come to share a more common set of beliefs about their work settings. Burke and Bennis (1961), for example, demonstrated that as group members continued to interact, they increased in consensus with one another, essentially moving toward greater perceptual congruity. Similarly, it is likely that as project members work together over a long period, they will reinforce their common views, commitments, and solution strategies. Such shared perceptions, created through group processes, act as powerful constraints on individual attitudes and behaviors. While these homogeneous tendencies isolate the group from many important outside information sources, they provide a great deal of assurance to group members. Given the certainty they facilitate, it is understandable that groups with shared systems of meaning and beliefs develop great stability and resistance to change.

In particular, the way engineering project groups come to view their outside work environments can be very critical. Given the low levels of external professional communication for the long-tenured project groups, for example, members may have come to believe that they possessed sufficient expertise and knowledge in their specialized technical areas so that it was no longer necessary to pay a great deal of attention to outside sources of new ideas and information. This perceptual outlook is known in the R&D community as the "Not Invented Here," or NIH syndrome. According to this syndrome, project members are more likely to see the virtue and superiority of their own ideas and activities while dismissing the suggestions and accomplishments of outside sources and competitors as inferior and weak.

The present findings are consistent with this NIH syndrome for long-tenured project groups. What is suggested by these results, then, is that communication patterns of project members and their subsequent effects on project performance might be strongly influenced and managed through staffing decisions. One would argue, for example, that the energizing and destabilizing function of new members should help prevent a group from developing behaviors and attitudes that coincide with the NIH syndrome. New group members not only have a relative advantage in generating fresh ideas and approaches, but through their active participation, project veterans might consider more carefully ideas and alternatives they might otherwise have ignored. In short, project newcomers represent

a novelty-enhancing condition, challenging and improving the scope of existing methods and accumulated knowledge.

Interestingly, R&D managers are usually not aware of the tenure demographics of their project groups. In the organization studied, none of the R&D managers could identify which projects had high levels of group longevity. In fact, they were very surprised that any of their projects had a mean group tenure of five or more years, especially since the facility was experiencing healthy growth. Moreover, conversations with managers from many high technology facilities has confirmed a similar lack of knowledge. As part of their project evaluation and technology planning activities, then, R&D managers might want to generate a more complete picture of their project groups' tenure distributions from which better staffing, hiring decisions, or reassignments of professionals among project activities can be made. While individual age and organizational tenure data are usually available through personnel files, individual project tenure and, more importantly, mean group tenure data are rarely part of a facility's ongoing information system.

Finally, whether project groups can circumvent the problem of low performance with high group longevity without rejuvenation from new members is an important question that needs to be addressed, especially in more stable R&D facilities. In the present site, none of the ten long-tenured projects was among the higher performing groups. All ten groups were rated as either average or below average in performance. As a result, it was not possible to determine from the present sample if high-performing, long-tenured project groups are able to maintain their effectiveness through appropriate high levels of communication and interaction with their more important communication domains. Additional research from other facilities is needed to ascertain just how consistent the current findings are with respect to project performance, group longevity, and project communications. In a general sense, we need to learn considerably more about the kinds of tasks, structures, and practices that are likely to prove useful in keeping a project group innovative and performing well as its members continue to work together.

Future Directions

This paper discusses only a few of the factors that might be important in developing a perspective about the issue of how the group is doing. Since most of the reported data was cross-sectional, one cannot really be sure of what happens to a project group as its members continue to interact. Longitudinal studies are clearly needed to verify the situation. In addition, communication patterns are only one of a large number of possible behavioral and perceptual changes that need to be investigated as a function of group longevity. For example, with increasing stability in membership, projects may become progressively committed to their particular solution strategies and methodologies, as well as increasingly narrow in their range of cognitive abilities.⁷ Similarly, there may develop strong tendencies to hire and attract only project members with similar beliefs, skills, and interests. Such tendencies toward homogeneity suggest even less diversity in the group, which would further inhibit group creativity and innovativeness.

These are just some of the issues that need to be addressed in order to build more useful and realistic frameworks for diagnosing group developments. Yet, in a more general sense, the challenge to the understanding, managing, and staffing of ongoing project groups lies in the ability of project members to maintain stability and continuity within the group, while retaining sufficient flexibility to keep abreast of external developments in the field. Further research is needed to learn more precisely what takes place in groups over time, so that appropriate action can be taken to mitigate the effects of group longevity on efficiency.

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